Six negative reflections in an orthogonal space of dimension five

Human Compression project: F001 working component

We treat one of the two involution types in odd dimension five. This argument does not exclude strings containing an involution whose negative eigenspace has dimension two. That remaining case is required for the full upper bound in Brooksbank's rank spectrum theorem. This draft has not received distinct review.

1 The classification input

Let V be a nondegenerate orthogonal space of dimension five over $\mathbf{F}_q$, where q is odd, and write B for its symmetric bilinear form. The reflection with root v, where $B(v,v) \neq 0$, is

$$s_v(x) = x - 2\frac{B(x,v)}{B(v,v)}v.$$

We retain the following part of the finite reflection classification. A finite irreducible orthogonal group in dimension five, generated by five reflections, is, after extending scalars and conjugating, either

1. $\mathrm{O}_5(q_0)$ or one of its subgroups generated by one reflection class, for a finite field of odd order q_0 of the same characteristic; or

2. a faithful modular reduction of the standard reflection group A_5, B_5, or D_5.

The groups in the first case contain $\Omega_5(q_0)$ and have a nondegenerate invariant form over $\mathbf{F}_{q_0}$. Allowing any q_0 avoids assuming that the originally chosen coefficient field is minimal. This is the dimension-five specialization of Monson–Schulte [1, Theorem 3.1], restated with its proof in Monson's author-hosted notes [2, Theorem 4.1, pp.15–17]. Those notes explicitly exclude the deleted symmetric module and the exceptional modular E_6 representation when the number of generating reflections is only the dimension. The underlying classification is due to Zalesskii–Serezhkin [3]. Its full proof remains an external acquisition cost.

We also retain the classical order formula

$$\omega(t) := |\Omega_5(t)| = \tfrac{1}{2}t^4(t^2-1)(t^4-1), \qquad |\mathrm{O}_5(t)| = 4\omega(t). \tag{1}$$

We retain also the simplicity of $\Omega_5(q)$ for odd $q \geq 5$. The homomorphism $\pi : \mathrm{O}(V) \longrightarrow \mathrm{SO}(V)$, $g \mapsto \det(g)g$, has kernel $\{1,-1\}$.

2 The end groups and their field of definition

Lemma 1. *Suppose $s_0,\dots,s_{r-1}$ are distinct reflections with roots $v_0,\dots,v_{r-1}$, commuting when their indices differ by more than one and not commuting at consecutive indices. Then $r \leq 6$. When $r = 6$, either consecutive set of five roots is a basis, and each corresponding end reflection group acts absolutely irreducibly.*

Proof. Distinct orthogonal reflections commute precisely when their roots are perpendicular. The Gram matrix is tridiagonal with nonzero entries immediately above and below the diagonal. A kernel vector is determined by its first coordinate: the first row determines the second coordinate, the second row the third, and so on. A nonzero kernel vector has both end coordinates nonzero, by applying this argument from either end. Thus its rank is at least $r - 1$ and at most five.

For $r = 6$ the rank is five. The roots span V and their one-dimensional relation space is the Gram kernel. No relation omitting an end root is possible, so either end five are a basis. Let H be one end group and extend scalars to an algebraic closure. If $0 \neq W \leq V$ is H-invariant, either every reflection fixes W pointwise, or some root lies in W, since $(s_i - 1)W \subseteq \langle v_i \rangle$. The first case places W in the perpendicular of a basis, which is zero. In the second, consecutive nonzero pairings propagate membership along the path, placing all five basis roots in W. Hence $W = V$. □

Lemma 2. *Suppose $r = 6$ above and $\langle -s_0, \ldots, -s_5 \rangle = \Omega_5(q)$. If $H = \langle s_0, \ldots, s_4 \rangle$ is of orthogonal type in the retained classification, then $\pi(H) = \Omega_5(q)$.*

Proof. Conjugate over an algebraic closure so that H and its invariant nondegenerate form B_0 are defined over $\mathbf{F}_{q_0}$. Absolute irreducibility shows that the original form B is a scalar multiple of B_0: the quotient of their matrix maps commutes with H. For $i \leq 4$, the negative eigenspace of s_i has a root defined over $\mathbf{F}_{q_0}$. The first four root lines therefore span a four-space defined over that field. Since s_5 commutes with $s_0, \ldots, s_3$, its root line is exactly the perpendicular of this four-space. It too is defined over $\mathbf{F}_{q_0}$, so s_5 is defined there and preserves B_0. Consequently

$$\langle s_0, \ldots, s_5 \rangle \leq \mathrm{O}_5(q_0) \tag{2}$$

after the same conjugation.

Since H contains $\Omega_5(q_0)$ and a reflection, $|H| \geq 2\omega(q_0)$. Also $\pi(H) \leq \Omega_5(q)$ and the kernel of π has order two, so $\omega(q_0) \leq \omega(q)$. The function $\omega(t)$ is strictly increasing for $t \geq 3$, whence $q_0 \leq q$. If $q_0 < q$, these are powers of the same odd prime, so $q \geq 3q_0$. Formula (1) gives $\omega(q) > 4\omega(q_0)$, contradicting (2) and the order of the signed image. Thus $q_0 = q$. Finally $|\pi(H)| \geq |H|/2 \geq \omega(q)$, giving the assertion. □

3 The finite alternatives

Lemma 3. *If an end group is a finite alternative, it has type A_5 or B_5. Its four consecutive product orders are respectively $(3, 3, 3, 3)$ or $(4, 3, 3, 3)$, up to reversal.*

Proof. In the standard A_5 representation, an involution is a reflection precisely when it is a transposition of the six coordinates. Its faithful irreducible reduction has characteristic different from two and three, and negative eigenspace dimensions are unchanged. Five transpositions generate S_6 only when their edges form a connected graph on six letters. With five edges it is a tree. Consecutive transpositions do not commute, so each consecutive product has order three.

In the signed permutation representation of B_5, the reflecting root lines are the short lines $\langle e_i \rangle$ and the long lines $\langle e_i \pm e_j \rangle$. This holds in every odd characteristic: an involutory signed permutation has one negative eigenvalue for each positive signed two-cycle and each negative fixed coordinate. The long reflections are precisely the reflections in D_5.

For five reflections to generate B_5, their permutation images must generate S_5. At least four are long. At least one is short, since long reflections lie in D_5. There are therefore exactly four long roots and one short root. The four long-root edges form a tree on the five coordinates. If the short root e_i were internal in the string, its two neighboring long roots would both involve i. Their edges are distinct and have only that coordinate in common, so those roots would not

be perpendicular, contradicting their distance two in the string. The short root is at an end. Its adjacent product has order four; adjacent long-root products have order three.

In D_5 every noncommuting pair of reflections has product order three. A connected string of five such reflections is a quotient of the Coxeter group $A_5 \cong S_6$, of order 720, and cannot generate D_5, of order 1920. □

Theorem 4. *Let $q \geq 5$ be odd. A string C-group marking of $\Omega_5(q)$ whose generators are all negatives of reflections has rank at most five. More strongly, any connected six-term string of distinct negative reflections generating $\Omega_5(q)$ has at least one end group equal to the whole group.*

Proof. The diagram of a C-marking of this simple group is connected: a disconnected string would give nontrivial commuting normal factors. Lemma 1 therefore bounds its rank by six. Suppose six generators exist and both signed end groups are proper. Lemma 2 excludes orthogonal type for either ordinary end group. Lemma 3 applies to both ends. Comparing their overlapping four-bond sequences shows that the middle three bonds are all three; each outer bond is either three or four.

Over an algebraic closure rescale the roots to have norm one. Put $x_i = B(v_{i-1}, v_i)^2$ for $1 \leq i \leq 5$. On the span of two independent roots the reflection product has determinant one and trace $4x_i - 2$. Hence order three gives $x_i = 1/4$, including characteristic three, and order four gives $x_i = 1/2$. If D_j is the leading j by j Gram determinant,

$$D_0 = D_1 = 1, \qquad D_{j+1} = D_j - x_j D_{j-1}. \tag{3}$$

The four possibilities give

$$\begin{array}{c|l} (3,3,3,3,3) & D_6 = 7/64 \\ (4,3,3,3,3),\ (3,3,3,3,4) & D_6 = 1/32 \\ (4,3,3,3,4) & D_6 = 0. \end{array} \tag{4}$$

As $D_6 = 0$, one outer four is impossible. The all-three case has characteristic seven and is a quotient of $A_6 \cong S_7$. Its signed image has order at most $5040 < \omega(q)$.

For the remaining case compare the following six root lines in the standard diagonal five-space:

$$e_1, \quad e_1 - e_2, \quad e_2 - e_3, \quad e_3 - e_4, \quad e_4 - e_5, \quad e_5. \tag{5}$$

After normalizing norms over the algebraic closure, their squared consecutive pairings agree with ours. Changing signs successively makes the Gram matrices identical. The first five roots form a basis with nondegenerate Gram determinant $1/16$. The isometry mapping these five normalized roots to (5) maps the sixth as well, since its pairings with a basis determine it. All six ordinary reflections are therefore conjugate to signed permutation matrices. Their group has order at most $|B_5| = 3840 < \omega(q)$, another contradiction.

Thus one signed end group is full. In a C-marking each generator is outside the subgroup on all the other marks, by the intersection property. Neither end can be full, excluding rank six. □

Scope and attribution. The reflection classification and classical order formula are retained results. This argument combines the field of definition of an end group with the overlapping finite end types. The proper signed S_7 counterexample to the source's arbitrary-subgroup lemma is consistent with this theorem. No bound for mixed involution types, complete F001 recovery, conjecture resolution, or priority is claimed.

References

[1] B. Monson and E. Schulte, *Reflection groups and polytopes over finite fields, I*, Advances in Applied Mathematics **33** (2004), 290–317, doi:10.1016/j.aam.2003.11.002.

[2] B. Monson, *Constructions of Symmetric Polytopes*, author-hosted course notes, supplied 54-page edition, `https://bmonson.ext.unb.ca/fields/outline.pdf`, Section 4, Theorem 4.1 and Table 3, pp.15–19.

[3] A. E. Zalesskii and V. N. Serezhkin, *Finite linear groups generated by reflections*, Mathematics of the USSR-Izvestiya **17** (1981), 477–503.